

Computer Programming Exercise Collection

Version 1.4 - Complete English Translation

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Preface. This collection of questions was written according to the needs of students. An effort has been made to present questions from a very simple level up to an advanced level. Therefore, the first questions in each section may seem very simple and repetitive, but they are necessary for beginner students. The final questions in each section may be very difficult and time-consuming, at the level of an Advanced Programming or Data Structures course; however, they are intended to be solved using ordinary programming knowledge, not knowledge from advanced courses or Data Structures. Please implement all questions so that you gain a complete practical understanding of programming. To make this collection more complete, please send new questions and your comments. With thanks - www.gClass.co - irani.gholamali@gmail.com

Translation note: the content, numbering, repeated questions, unusual score signs, filenames, and apparent source typos are preserved rather than silently corrected. Where a source typo is especially likely to confuse the reader, it is marked as written in the source.

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Section 1 - Questions Related to if, for, and while

1. Write a program that reads an integer from 1 to 365 and prints the current month.
2. Write a program that reads an integer from 1 to 365 and prints the current day of the month.
3. Write a program that reads n as the amount of money requested from an ATM, then breaks n into 1-, 5-, 10-, and 50-thousand-toman denominations.
4. Can you find and print the absolute value of an input number without using an if statement?
5. Can you find and print the minimum of two input numbers without using an if statement?
6. Can you find and print the maximum of two input numbers without using an if statement?
7. Write a program that prints the numbers from 1 to 100.
8. Write a program that prints the sum of the numbers from 1 to 100.
9. Write a program that prints the sum of the even numbers from 1 to 100.
10. Write a program that prints the sum of the odd numbers from 1 to 100.
11. Write a program that reads a number and then prints the natural numbers less than or equal to it.
12. Write a program that reads a number and then prints its divisors.
13. Write a program that reads a number and then prints its odd divisors.
14. Write a program that reads a number and then prints its even divisors.
15. Write a program that reads a number and then prints the sum of its divisors.
16. Write a program that reads a number and then prints the number of its divisors.
17. Write a program that reads a number and then prints the sum of its odd divisors.
18. Write a program that reads a number and then prints the number of its odd divisors.
19. Write a program that reads a number and then prints the sum of its even divisors.
20. Write a program that reads a number and then prints the number of its even divisors.
21. Write a program that reads a number and then prints the average of its divisors.
22. Write a program that reads a number and then prints the average of its even divisors.
23. Write a program that reads two numbers, then calculates and prints their GCD and LCM.
24. Write a program that reads a number and prints "Yes" if it is prime; otherwise print "No".
25. Write a program that reads a number, calculates its factorial, and prints it.
26. Write a program that reads 10 numbers, calculates their average, and prints it.
27. Write a program that reads 100 numbers and prints their minimum.
28. Write a program that reads 100 numbers and prints their maximum.
29. Write a program that reads 100 numbers and prints their second minimum.
30. Write a program that reads 100 numbers and prints their second maximum.
31. Write a program that prints all prime numbers less than 1000.
32. Write a program that reads a number and prints its two-digit divisors.
33. Write a program that reads a number and prints the sum of its two-digit divisors.
34. Write a program that reads a number and prints the number of its two-digit divisors.
35. Write a program that reads a number and prints the average of its two-digit divisors.
36. Write a program that reads 100 numbers and prints the number of prime numbers among them.
37. Write a program that reads 100 numbers and prints the sum of the prime numbers among them.
38. Write a program that reads 100 numbers and prints the number of odd numbers among them.
39. Write a program that reads 100 numbers and prints the number that has the greatest number of divisors.
40. Write a program that prints the 50th prime number.
41. Write a program that prints the first 50 terms of the Fibonacci sequence.
42. Write a program that reads a number and prints the sum of its digits.

43. Write a program that reads a number and prints the sum of its even digits.
44. Write a program that reads a number and prints the sum of its odd digits.
45. Write a program that reads a number and prints the number of its digits.
46. Write a program that reads a number and prints the number of its even digits.
47. Write a program that reads a number and prints the number of its odd digits.
48. Write a program that reads a number and prints the product of its digits.
49. Write a program that reads a number and prints the product of its even digits.
50. Write a program that reads a number and prints the product of its odd digits.
51. Write a program that reads a number and prints the average of its digits.
52. Write a program that reads a number and prints the average of its even digits.
53. Write a program that reads a number and prints the average of its odd digits.
54. Write a program that reads a number and prints its reverse.
55. Write a program that reads a number and prints the reverse formed by its odd digits.
56. Write a program that reads a number and prints the reverse formed by its even digits.
57. Write a program that reads two numbers and prints their greatest common divisor.
58. Write a program that reads two numbers and prints their least common multiple.
59. Write a program that reads a number and factorizes it into prime factors.
60. Write a program that prints the 20th prime number in the Fibonacci sequence.
61. Write a program that prints Fibonacci terms that are prime, up to 50 such terms.
62. Write a program that reads a number and determines whether it occurs in the Fibonacci sequence.
63. Write a program that reads 100 numbers and prints how many of them occur in the Fibonacci sequence.
64. Write a program that reads 100 numbers and prints the sum of all digits of those 100 numbers.
65. Write a program that reads 100 numbers and prints the total number of digits in those 100 numbers.
66. Write a program that reads 100 numbers and prints the average of their digit-averages.
67. Write a program that reads 100 numbers and prints how many of them are equal to their reverses.
68. Write a program that reads a number and removes its zero digits.
69. Write a program that reads a number and prints the number of its prime divisors.
70. Write a program that reads a number and prints the sum of its prime divisors.
71. Write a program that reads a number and prints the average of its prime divisors.
72. Write a program that prints the first Fibonacci number that is equal to its reverse.
73. Write a program that reads n and prints n rows of each of the six star patterns shown below.

Pattern 1:

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Pattern 2:

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Pattern 3:

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Pattern 4:

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*
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Pattern 5:

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*
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Pattern 6:

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*
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**
*
```

- 74.** Write a program that prints the multiplication table.
- 75.** Write a program that prints the multiplication table in base 8.
- 76.** Write a program that first reads 100 numbers from the user, then prints the number of values that are both prime and present in the Fibonacci sequence.
- 77.** Write a program that reads a number and determines whether it is a perfect number.
- 78.** Write a program that prints the perfect numbers less than 1000.
- 79.** Write a program that reads 100 numbers and prints the number of values that are perfect numbers.
- 80.** Write a program that reads 100 numbers and prints the number of values that are both perfect and divisible by 2.
- 81.** Write a program that reads a number and removes its zero digits.
- 82.** Write a program that reads a number and prints its equivalent in base 2.
- 83.** Write a program that reads a number and prints its equivalent in base 8.
- 84.** Write a program that reads two numbers in base 8, then calculates and prints their sum and product.
- 85.** Write a program that reads a number in base 2 and prints its equivalent in base 10.
- 86.** Write a program that reads a number in base 8 and prints its equivalent in base 10.
- 87.** Write a program that reads a number in base 2 and converts it to base 8.
- 88.** Write a program that reads a number and prints its equivalent in base 16.
- 89.** Write a program that reads a number in base 2 and converts it directly to base 16.
- 90.** For a specific equation, write a program that reads x_1 and x_2 and computes the integral from x_1 to x_2 with accuracy 10^{-7} , then prints the result. Use the rectangular, trapezoidal, Simpson, and midpoint methods.

91. Write a program that computes and prints the following integral with accuracy 0.00001: integral from 1 to 5 of $\sin(x)/x \, dx$.
92. For a specific nonlinear equation, write a program that reads x_1 and x_2 such that $f(x_1) \cdot f(x_2) < 0$ and prints the root in this interval. Use the bisection, false-position/correction, secant, Newton, and fixed-point methods.
93. Write a program that calculates and prints all roots of a specific nonlinear equation.
94. Write a program that reads m and n , then calculates and prints the binomial coefficient $C(m, n)$.
95. Write a program that reads n and prints the n th row of Pascal's triangle.
96. Write a program that reads n and prints n rows of Pascal's triangle.
97. Read a fractional number in base 10 and print its equivalent in base 2, to 5 fractional digits.
98. Read a fractional number in base 10 and print its equivalent in base 8, to 8 fractional digits.
99. Read a fractional number in base 2 and print its equivalent in base 10, to 5 fractional digits.
100. Read a fractional number in base 10 and print its equivalent in base 16, to 8 fractional digits.
101. Write a program that prints all three-digit numbers that can be written as $xyz = x^3 + y^3 + z^3$. For example: $153 = 1^3 + 5^3 + 3^3$.
102. Evaluate the series: $1 + 2 + 3 + \dots + 100$.
103. Evaluate the series: $1 - 2 + 3 - 4 + \dots - 100$.
104. Evaluate the series: $1 - 3 + 5 - 7 + \dots - 99$.
105. Evaluate the series: $1/1 + 1/2 + 1/3 + \dots + 1/100$.
106. Evaluate the series: $1/1! + 1/2! + 1/3! + \dots + 1/100!$.
107. Evaluate the series: $1/20 + 2/19 + 3/18 + \dots + 20/1$.
108. Evaluate the series: $1/1! - 1/2! + 1/3! - \dots - 1/100!$.
109. Evaluate the series: $1/1! + 1/3! + 1/5! + \dots + 1/99!$.
110. Evaluate the series: $1 - 1/2! + 1/4! - \dots - 1/100!$.
111. Evaluate the infinite series $1/1! + 1/2! + 1/3! + \dots$ with accuracy 10^{-7} .
112. Read x , then evaluate the series $1 + x/1! + x^2/2! + x^3/3! + \dots$ with accuracy 10^{-7} .
113. Read x , then evaluate the series $1 + x^2/2! + x^4/4! + \dots$ with accuracy 10^{-7} .
114. Read x , then evaluate the series $x - x^3/3! + x^5/5! - \dots$ with accuracy 10^{-7} .
115. Evaluate the series: $1!/1^1 + 2!/2^2 + 3!/3^3 + \dots + 100!/100^{100}$.
116. Evaluate the infinite series $1!/1^1 + 2!/2^2 + 3!/3^3 + 4!/4^4 + \dots$ with accuracy 10^{-7} .
117. Evaluate the infinite series $1!/1^1 - 3!/3^3 + 5!/5^5 - 7!/7^7 + \dots$ with accuracy 10^{-10} .
118. Evaluate the continued fraction $1 + 1/(2 + 1/(2 + 1/(2 + 1/(2 + \dots))))$ with accuracy 10^{-10} .
119. Evaluate the series $\ln(x) = 2[(x-1)/(x+1) + (1/3)((x-1)/(x+1))^3 + (1/5)((x-1)/(x+1))^5 + \dots]$ with accuracy 10^{-10} .
120. Find the Taylor-series expansions related to $\sin(x)$, $\cos(x)$ [written this way in the source], e^x , $\sqrt{x}$, $\ln(x)$, etc. Write a program that reads x and evaluates the selected series with accuracy 10^{-7} . The source suggests consulting the Wikipedia page on Taylor series for the formulas.

Section 2 - Questions Related to One-Dimensional Arrays

121. Write a program that reads 5 numbers and prints them from last to first.
122. Write a program that reads 20 numbers and prints them from last to first.
123. Write a program that reads 20 numbers and prints the numbers that are greater than the average of the 20 entered numbers.
124. Write a program that reads 20 numbers and prints how many of them are greater than their average.
125. Write a program that reads 20 numbers and prints the values that are prime and greater than the average.
126. Write a program that reads 20 numbers and prints the values that occur in the Fibonacci sequence and are greater than the average.
127. Write a program that reads n numbers and prints them from last to first.
128. Write a program that reads n numbers and then a number m, and prints the number of occurrences of m among the n numbers.
129. Write a program that reads n numbers and then a number m, and prints the positions of m among the n numbers.
130. Write a program that reads a long integer and prints its equivalent in base 2.
131. Write a program that reads 20 numbers and removes the value in the 5th position by shifting the elements and reducing the total element count, then prints the remaining values.
132. Write a program that reads n numbers and then a number m, removes all occurrences of m among the n numbers, and prints the remaining values.
133. Write a program that reads n numbers and positions p1 and p2, removes the elements from positions p1 through p2, and prints the remaining values.
134. Write a program that reads 10 numbers, circularly shifts them 3 positions forward, and prints the result.
135. Write a program that reads n numbers and then m, circularly shifts the n numbers m positions forward, and prints the result.
136. Write a program that reads n numbers into an array, reverses the array around its midpoint (first with last, second with next-to-last, etc.), and prints the resulting array.
137. Write a program that reads n numbers into an array, stores the positive and negative values in separate arrays, and prints each result array.
138. Write a program that reads n numbers into an array, stores the prime and non-prime values in separate arrays, and prints each result array.
139. Write a program that reads n numbers into an array, stores the prime, composite, and Fibonacci values in separate arrays, and prints each result array.
140. Write a program that reads n numbers, removes the even values, and prints the final values.
141. Write a program that reads n numbers, removes the prime values, and prints the final values.
142. Write a program that reads n numbers, then reads m and k, and inserts m into position k of the array.
143. Write a program that reads n numbers, places the number of divisors of each number in the next array position after that number, and prints all values in the final array.
144. Write a program that reads n numbers, places the number of digits of each number in the next array position after that number, and prints all values in the final array.
145. Write a program that reads n numbers and then m additional numbers, finds the common numbers in the two lists, and prints them.
146. Write a program that reads n numbers and then m, and prints the number of occurrences of m among the n numbers.
147. Write a program that reads n numbers and places the frequency of each number in another corresponding array.
148. Write a program that reads n numbers and prints the number with the greatest frequency.
149. Write a program that reads n numbers and prints the index of an array cell whose value has the greatest frequency.
150. Write a program that reads n numbers, removes duplicate values, and prints the result.
151. Write a program that reads n numbers and prints all values that have the greatest frequency.
152. Write a program that prints a random number.

153. Write a program that prints a random integer between 1 and 100.
154. Write a program that prints a random integer between a and b.
155. Write a program that generates 20 random numbers between 1 and 100, stores them in an array, and prints the array.
156. Write a program that generates and prints 20 non-repeating random numbers between 1 and 100.
157. Write a program that reads n numbers between 1 and 10, then prints the frequency of each value.
158. Using one-dimensional arrays, write a program that reads n and prints the nth row of Pascal's triangle.
159. Using one-dimensional arrays, write a program that reads n and prints n rows of Pascal's triangle.
160. Write a program that reads n numbers and determines whether they are sorted.
161. Write a program that reads n numbers and prints their median.
162. Write a program that reads n numbers and sorts them using bubble sort, insertion sort, and selection sort.
163. Write a program that reads two sorted groups of m and n numbers, merges them into a third sorted array, and prints the result. (Merge two sorted arrays.)
164. Write a program that reads n sorted numbers and then m, and searches for m among the n numbers using binary search.
165. Write a program that reads n sorted numbers and then m, and searches for m among the n numbers using ternary search.
166. Write a program that calculates and prints 1000!.
167. Write a program that calculates and prints 100000!.
168. Write a program that calculates and prints 1000000!.
169. Write a program that calculates and prints 2^{100} .
170. Write a program that reads two long integers m and n, calculates m^n , and prints it.
171. Write a program that calculates and prints $(1000!)^3$.
172. Write a program that reads two large integers (for example, 100-digit integers) and prints their sum.
173. Write a program that reads two large integers (for example, 100-digit integers) and prints their difference.
174. Write a program that reads two large integers (for example, 100-digit integers) and prints their product.
175. Write a program that reads two large integers (for example, 100-digit integers) and prints their integer quotient.
176. Write a program that reads two large integers (for example, 100-digit integers) and prints their integer remainder.
177. Write a program that reads two large integers (for example, 100-digit integers), calculates their decimal quotient, and prints it.
178. Evaluate the series: $1^1 + 2^2 + 3^3 + \dots + 100^{100}$.
179. Evaluate the series: $1! + 2! + 3! + \dots + 100!$.
180. Write a program that calculates and prints $(2^{64})!$.

Section 3 - Questions Related to Strings

181. Write a program that reads a letter and displays it in uppercase (for example, print A instead of a).
182. Write a program that reads a letter and displays it in lowercase (for example, print a instead of A).
183. Write a program that reads a character and prints its ASCII code.
184. Write a program that prints all ASCII codes from 0 to 255 and their corresponding characters in table form.
185. Write a program that reads a number and prints its equivalent in base 16.
186. Write a program that reads a string and prints it.
187. Write a program that reads a number in base 16 and prints its equivalent in base 10.
188. Write a program that reads a string, converts all uppercase letters to lowercase and all lowercase letters to uppercase, and prints the result.
189. Write a program that reads a string and prints the average of the digit characters contained in it.
190. Write a program that reads a string, removes all digit characters from it, and prints the result.
191. Write a program that reads string S and character c and prints the positions where c occurs in S.
192. Write a program that reads strings S1 and S2, finds the number of occurrences of S2 in S1, and prints it.
193. Write a program that reads a string, converts the first letter of every word to uppercase, and prints the result.
194. strtoint: Write a program that reads a number of at most 10 digits as a string, obtains its numeric value from the string, and prints it. Example: "538" => 538.
195. charAt: Write a program that reads a string and a number n and prints the character at position n.
196. Reverse: Write a program that reads a string and prints it from last to first.
197. ToLower: Write a program that reads a string, converts all of its letters to lowercase, and prints it.
198. ToUpper: Write a program that reads a string, converts all of its letters to uppercase, and prints it.
199. SetW: Write a program that reads a string and then a number n, inserts space characters at the beginning of the string until its length reaches n, and prints the result.
200. SubString: Write a program that reads a string S and two numbers n1 and n2, copies positions n1 through n2 from S into a new string, and prints the result.
201. Ltrim: Write a program that reads a string and removes the spaces on its left side.
202. Rtrim: Write a program that reads a string and removes the spaces on its right side.
203. Trim: Write a program that reads a string and removes the spaces at its beginning and end.
204. FullTrim: Write a program that reads a string, removes the spaces at its beginning and end, reduces every internal run of more than one space to a single space, and prints the result.
205. IndexOf: Write a program that reads a string S, a character c, and a number n, and from position n onward prints the first occurrence position of c in S. Print -1 if it is not found.
206. IndexOf2: Write a program that reads string S1, another string S2, and a number n, and from position n onward in S1 finds and prints the first occurrence position of S2. Print -1 if it is not found.
207. Compare: Write a program that reads two strings, compares them for equality, and prints the result.
208. Compare2: Write a program that reads two strings, compares them for equality without considering uppercase/lowercase differences, and prints the result.
209. Remove: Write a program that reads a string and a character and removes the specified character from the string.
210. Remove2: Write a program that reads strings S1 and S2 and removes all occurrences of S2 from S1.
211. Replace: Write a program that reads strings S1, S2, and S3 and replaces every occurrence of S2 inside S1 with S3.
212. Split: Write a program that reads a string S and a character c, splits S into multiple strings based on occurrences of c, and prints the resulting strings.
213. Write a program that reads a string and prints all of its subsets.
214. Write a program that reads a string of length n and prints its m-element subsets ($m < n$).
215. Write a program that reads strings S1 and S2 and prints the number of occurrences of S2 in S1.
216. Write a program that reads a string and prints its repeated characters.

217. Write a program that reads a string and prints its repeated sentences.
218. Write a program that reads a string and prints its complement with respect to the reference set of all English letters.
219. Write a program that reads two strings S1 and S2 and prints their Cartesian product.
220. Write a program that reads two strings S1 and S2 and prints the characters common to both.
221. Write a program that reads two strings S1 and S2 and prints the intersection of the sentences in the two strings.
222. Write a program that reads a string and prints all of its permutations.
223. Write a program that reads 5 strings and prints the permutations of the entered strings together.
224. First define a reversible encryption algorithm. Then write a program that reads a string and encodes it, or reads an encoded string and decodes it.
225. Write a program that reads an integer of at most 9 digits and prints its Persian equivalent in words.
226. Write a program that reads an integer of at most 9 digits and prints its English equivalent in words.
227. Write a program that reads a decimal number with at most 9 integer digits and 8 fractional digits and prints its Persian equivalent in words.
228. Write a program that reads two very large integers as strings, calculates their sum, and prints it.
229. Write a program that reads two very large integers as strings and prints their difference.
230. Write a program that reads two very large integers as strings and prints their product.
231. Write a program that reads two very large integers as strings and prints their integer quotient.
232. Write a program that reads two very large integers as strings and prints their integer remainder.
233. Evaluate the series: $1^1 + 2^2 + 3^3 + \dots + 100^{100}$.
234. Evaluate the series: $1! + 2! + 3! + \dots + 100!$.
235. Write a program that calculates and prints $(2^{64})!$.
236. Write a program that reads two polynomials as strings and prints their sum.
237. Write a program that reads two polynomials as strings and prints their difference.
238. Write a program that reads two polynomials as strings and prints their product.
239. Write a program that reads two polynomials as strings, calculates their quotient and remainder, and prints them.
240. Write a program that reads a polynomial as a string and prints its derivative.
241. Write a program that reads a polynomial as a string and prints its integral.
242. Write a program that reads a mathematical expression containing digits and the + sign, such as "7 + 534 + 12", as a string, calculates its numeric value, and prints it.
243. Do the previous problem again with the - sign added.
244. Write a program that reads a mathematical expression containing digits, +, and multiplication, such as "173 + 34 * 51 + 12", as a string, calculates its numeric value, and prints it.
245. Write a program that reads a mathematical expression containing digits, +, multiplication, and parentheses as a string, calculates its numeric value, and prints it.
246. Write the arithmetic-expression problems for decimal values and negative numbers as well.
247. Write a program that reads a number and prints its equivalent in base 32.
248. Write a program that reads a sentence, capitalizes the first letter of every word, and prints the result.
249. Write a program that reads a sentence and removes repeated words.
250. Write a program that reads a mathematical equation and draws its graph.

Section 4 - Questions Related to Two-Dimensional Arrays

251. Write a program that reads a 10 x 10 matrix from the user.
252. Write a program that reads a 10 x 10 matrix and prints it.
253. Write a program that reads m and n, generates an identity (unit) matrix, and prints it.
254. Write a program that reads a 10 x 10 matrix and prints only its main diagonal.
255. Write a program that reads a 10 x 10 matrix and prints only its secondary diagonal.
256. Write a program that reads a 10 x 10 matrix and prints the average of the values on its main diagonal.
257. Write a program that reads a 10 x 10 matrix and prints only its even-numbered rows.
258. Write a program that reads a 10 x 10 matrix and prints the maximum value in it.
259. Write a program that reads a 10 x 10 matrix and prints the maximum value of each row.
260. Write a program that reads a 10 x 10 matrix and prints the minimum value of each column.
261. Write a program that reads a 10 x 10 matrix and two numbers m and n, swaps rows m and n, and prints the resulting matrix.
262. Write a program that reads a 10 x 10 matrix and two numbers m and n, swaps columns m and n, and prints the resulting matrix.
263. Write a program that reads m and n and then an n x m matrix, stores its transpose in another matrix, and prints it.
264. Write a program that reads m and n and then an n x m matrix, sets the lower triangle of the matrix to zero, and prints the result.
265. Write a program that reads m and n and then an n x m matrix, sets the upper triangle of the matrix to zero, and prints the result.
266. Write a program that reads m and n and then an n x m matrix, then reads a value P and prints the coordinates of every occurrence of P in the matrix.
267. Write a program that reads 100 numbers into a 100-element array, then copies the 100 values row by row into a 10 x 10 matrix.
268. Write a program that reads 45 numbers into a 45-element array, then copies the 45 values row by row into a 5 x 9 matrix.
269. Write a program that reads 45 numbers into a 45-element array, then copies the 45 values row by row into a 9 x 5 matrix.
270. Write a program that reads 45 numbers into a 45-element array, then copies the 45 values column by column into a 5 x 9 matrix.
271. Write a program that reads 45 numbers into a 45-element array, then copies the 45 values column by column into a 9 x 5 matrix.
272. Write a program that reads m and n and then an n x m matrix, sorts each row of the matrix, and prints the result.
273. Write a program that reads m and n and then an n x m matrix, sorts each column of the matrix, and prints the result.
274. Write a program that reads m and n and then two n x m matrices, calculates their sum into a third matrix, and prints the result.
275. Write a program that reads m and n and then an n x m matrix, then reads P, adds P to every matrix element, and prints the final matrix.
276. Write a program that reads m and n and then an n x m matrix, then reads P, multiplies every matrix element by P, and prints the final matrix.
277. Write a program that reads m, n, and k, then reads two matrices n x k and k x m, multiplies them, and prints the resulting n x m matrix.
278. Write a program that reads n and then an n x n matrix, calculates its inverse, and prints it.
279. Write a program that reads n and then an n x (n+1) augmented matrix as the coefficients of an n x n linear system, calculates the n roots, and prints them. Use Gauss, Gauss-Jordan, and LU methods.
280. Write a program that reads a matrix, calculates its eigenvalues, and prints them.

Section 5 - Questions Related to Methods

- 281.** A method for returning the min of two int numbers: `public static int max(int x, int y);` [Method name/signature preserved exactly as written in the source.]
- 282.** A method for returning the min of two float numbers: `public static float max(float x, float y);` [Method name/signature preserved exactly as written in the source.]
- 283.** A method for the absolute value of one int number: `public static int abs(int x);`
- 284.** A method for the absolute value of one float number: `public static float abs(float x);`
- 285.** Most programs in Section 1 (if, for, while) can also be implemented using methods. For example, instead of "Write a program that reads a number and prints the number of its divisors," write: "Write a method that receives a number as a method input, calculates the number of its divisors, and returns that count as the method output."
- 286.** Most programs in Section 2 (one-dimensional arrays) can also be implemented using methods. For example, the merge-two-sorted-arrays problem can be stated as: "Write a method that receives two sorted arrays as method inputs, places their sorted merge in a third array, and returns it as the method output."
- 287.** Likewise, most questions in Sections 1, 2, 3, and 4 can be expressed as methods. Several examples follow.
- 288.** Write a method that receives a string, converts its lowercase letters to uppercase and its uppercase letters to lowercase, and returns the result.
- 289.** Write a method that receives strings S1 and S2, finds and removes every S2 inside S1, and returns the result.
- 290.** Write a method that receives a string, converts the first letter of every word to uppercase, and returns the result.
- 291.** Write a method that receives a string, separates its words, and returns them as an array of strings.
- 292.** Write a method that receives n ($1 \leq n \leq 1000$) and an array, calculates $n!$, places the result inside the array, and returns the length of the resulting array.
- 293.** Write a method that receives three arrays together with the lengths of the first two arrays, calculates the product of the first two arrays, places the result in the third array, and returns the length of the result array.
- 294.** Write a method that receives a string containing an ordinary decimal number such as "-328.098" or "101.009", converts the decimal number in the string to a decimal numeric value, and returns it.
- 295.** BigSum: a method that receives two positive large numbers as strings and returns their sum as a string.
- 296.** BigMinus: a method that receives two positive large numbers as strings, where the first number is larger, and returns their difference as a string.
- 297.** BigMultiple: a method that receives two positive large numbers as strings and returns their product as a string.
- 298.** BigDiv: a method that receives two large numbers as strings and returns their integer quotient as a string.
- 299.** BigMod: a method that receives two large numbers as strings and returns the remainder of their integer division as a string.
- 300.** BigCompare: a method that receives two positive large numbers as strings and returns the comparison result as 1, -1, or 0.
- 301.** PSum: a method that receives two polynomials as strings and returns their sum as a string.
- 302.** PMinus: a method that receives two polynomials as strings and returns their difference as a string.
- 303.** PMultiple: a method that receives two polynomials as strings and returns their product as a string.
- 304.** PDiv: a method that receives two polynomials as strings and returns their quotient as a string.
- 305.** PMod: a method that receives two polynomials as strings and returns the remainder of their division as a string.
- 306.** Write a method named MLPlusMinus that receives a mathematical expression as a string containing positive or negative long integers and the - and + operators, and returns the result as a long. Examples: "-438-2+23" or "329+9-9128".
- 307.** Write a method that receives a mathematical expression as a string containing positive or negative long integers and the operators -, +, *, and /, and returns the result as a double.
- 308.** Write a method that receives a mathematical expression as a string containing positive or negative long integers and the operators -, +, *, /, and ^, and returns the result as a double.
- 309.** Write a method that receives a mathematical expression as a string containing positive or negative long integers and the operators -, +, *, /, ^, and parentheses, and returns the result as a double.

310. Write a method that receives a mathematical expression as a string containing positive or negative decimal (double) numbers; the operators -, +, *, /, ^; parentheses; and functions such as Exp, Sqr, Ln, Log, Tan, Cos, and Sin; and returns the result as a double.

311. All String-related methods: StrToInt, CharAt, Reverse, ToLower, ToUpper, SetW, SubString, LTrim, RTrim, Trim, FullTrim, IndexOf, Compare, Remove, Replace, Split.

Section 6 - Sample Final Project Lists

First Semester 1389

Numerical Integration Project [Score: 1]

The project must be a single program that calculates and prints the integral over a specified interval using all 7 methods below. Using functions for each method is mandatory. Methods: rectangular, trapezoidal, Simpson, midpoint, Romberg, 2-point Gaussian, and 3-point Gaussian.

Polynomial Roots Calculation Project [Score: 1]

Calculate and print the roots of a polynomial.

Linear Systems Project [Score: 2]

Receive an $n \times (n+1)$ matrix and calculate the solution of the system using the following 4 methods. You may write a separate program for each method. Methods: Gauss, Gauss-Jordan, Gaussian pivoting, and LU decomposition.

Nonlinear Equations Project [Score: 1]

For a nonlinear equation already present in the program, calculate its zeros using 5 methods after receiving the relevant interval and calculation precision. Using functions for each method is mandatory. Methods: bisection, false-position/correction, fixed point, Newton, and secant.

Matrix Inverse Project [Score: 1]

Read an $n \times n$ matrix, calculate its inverse, and print it.

Big Numbers Calculator Project [Score: 2]

1) Read two large numbers and an operator from *, -, + and print the result. 2) Read two large numbers and print their quotient. 3) For an ordinary number, calculate and print its factorial. 4) Read two ordinary numbers and print the result of exponentiation. The entire project must be one program, and using functions for each operation is mandatory.

Calculate and Store Large Factorials [Score: 2]

Calculate $100000!$, $200000!$, $300000!$, ..., $1000000!$ and store them in separate files. Then read a number and, by reading the nearest stored value from a file, calculate its factorial.

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First Semester 1390

Numerical Analysis Project (NAP) - P11 [Score: 1]

Numerical integration using rectangular, trapezoidal, Simpson, and midpoint methods: for a specified equation defined inside the program as a function, read the other specifications such as the first point, last point, precision, etc., and print its definite integral over the relevant interval. The program must be menu-driven. One menu item must print the results of all methods together.

Numerical Analysis Project (NAP) - P12 [Score: 1]

Solve nonlinear equations using false-position/correction, fixed-point, Newton, and secant methods, in the same manner as Part 1.

Numerical Analysis Project (NAP) - P13 [Score: 4]

Solve linear systems using Gauss and Gauss-Jordan methods: read an $n \times (n+1)$ matrix as a linear system, solve it using Gauss or Gauss-Jordan, and print the results.

Numerical Analysis Project (NAP) - P14 [Score: 4]

Matrix operations including addition and multiplication of two matrices and inversion of one matrix: use a menu to read the two matrices for addition/multiplication and perform the selected operation; for inversion, read one matrix and print its inverse.

Big Numbers Calculator (BNC) - P21 [Score: 4]

Addition, subtraction, and multiplication of two large numbers: read two large numbers and perform the corresponding operations.

Big Numbers Calculator (BNC) - P22 [Score: 15]

Division: read two large numbers, perform integer division, and print the quotient and remainder. This program must use functions for addition, subtraction, multiplication, comparison, etc.

Big Numbers Calculator (BNC) - P23 [Score: 1]

Power and factorial of large numbers: same style as Part 1.

Equation Lab (EL) - P31 [Score: 3]

Addition, subtraction, and multiplication of polynomials: read two polynomials and print their sum, difference, and product.

Equation Lab (EL) - P32 [Score: 5]

Polynomial division: read two polynomials, divide them, and print the quotient and remainder.

Equation Lab (EL) - P33 [Score: 2]

Derivative and integral of polynomials: read one polynomial, calculate its derivative and integral, and print them.

Math Lab (ML) - P41 [Score: 5]

Simple mathematical expressions: read as a string an expression containing numbers and - and +, calculate its value, and print it.

Math Lab (ML) - P42 [Score: 5]

Ordinary mathematical expressions: read as a string an expression containing numbers and -, +, /, *, calculate its value, and print it.

Math Lab (ML) - P43 [Score: 5]

Parenthesized mathematical expressions: read as a string an expression containing numbers and -, +, /, *, (,), calculate its value, and print it.

Math Lab (ML) - P44 [Score: 15]

Evaluation of mathematical functions: read a mathematical function as a string containing multi-digit numbers, -, /, *, +, ^, parentheses, and functions such as sin, cos, tan, log (same as Ln), sqr (square root), and exp. Example: $\text{Log}(x \cdot \sin(2x)) - \exp(x^2 - 1) + 2x^{12} - 10x + \sqrt{x} \cdot x^3$. Then read a number such as 2.5, substitute it into the function, calculate the result, and print it.

University System (US) - P51 [Score: 30]

Student and course information registration system. Required features: 1) Add student data including student number, first name, last name, birth-certificate number, and address; all student data must be stored in a file sorted by student number. 2) Edit, delete, and perform simple and advanced searches on all student data. 3) Add course data including course code, course name, number of theoretical units, and number of practical units. 4) Edit, delete, and perform simple and advanced searches on all course data. 5) Register, edit, and search each student's course selection.

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First Semester 1391

Numerical Analysis Project (NAP) - P11 [Score: 10]

Solve linear systems using Gauss and Gauss-Jordan: read an $n \times (n+1)$ matrix as a linear system, solve it using Gauss and Gauss-Jordan, and print the results.

Big Numbers Calculator (BNC) - P21 [Score: 3]

Addition, subtraction, and multiplication of two large numbers: read two large numbers and perform the corresponding operations.

Big Numbers Calculator (BNC) - P22 [Score: 6]

Division: read two large numbers, perform integer division, and print the quotient and remainder. Use functions for addition, subtraction, multiplication, comparison, etc.

Big Numbers Calculator (BNC) - P23 [Score: 1]

Power and factorial of large numbers: same style as Part 1.

Polynomial Lab (PL) - P31 [Score: 3]

Read two polynomials and print their sum, difference, and product.

Polynomial Lab (PL) - P32 [Score: 5]

Read two polynomials, divide them, and print the quotient and remainder.

Polynomial Lab (PL) - P33 [Score: 2]

Read one polynomial, calculate its derivative and integral, and print them.

Math Lab (ML) - P41 [Score: 5]

Read a parenthesized mathematical expression containing numbers and -, +, /, *, (,) as a string, calculate its value, and print it.

Math Lab (ML) - P42 [Score: 25]

Read a string as a mathematical function containing digits, -, /, *, +, ^, parentheses, and functions such as sin, cos, tan, log (same as Ln), sqr (square root), and exp. Example: $\text{Log}(x \cdot \sin(2x)) - \exp(x^2 - 1) + 2x^{12} - 10x + \sqrt{x} \cdot x^3$. Then read a number such as 2.5, substitute it into the function, calculate the result, and print it.

Math Lab (ML) - P43 [Score: 10]

Read a mathematical function in the same way as P42 and print all zeros of the function.

Library System (LS) - P51 [Score: 35]

University library information system. Required features: 1) Add, edit, delete, and perform simple and advanced searches on student information including student number, first name, last name, national ID, phone number, and address; store all student data in a file sorted by student number. (10 marks) 2) Add, edit, delete, and perform simple and advanced searches on book information including book number, title, author, translator, and number of copies; store all book data in a file sorted by book number. (10 marks) 3) Register and search book-loan and book-return information. (15 marks)

Very Big (VB) - P61 [Score: +20]

Calculate $(2^{64})!$ and store it in a file.

Quick Fact (QF) - P71 [Score: 5]

Calculate $100000!$, $200000!$, ..., $1000000!$ and store them in separate files.

Quick Fact (QF) - P72 [Score: 10]

Read a number and, by reading the nearest stored value from a file, calculate its factorial and store it in a file named with the input number.

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First Semester 1392**P1 Big Numbers (BN) - P11** [Score: 5]

First implement BigSum, BigMinus (first number larger), BigMultiple, and BigCompare (returns 1, -1, or 0). Based on those methods and any additional methods you define, implement BigDiv, which returns the integer quotient as a string (5 marks), and BigMod, which returns the integer-division remainder as a string (-5 marks), and demonstrate their use with an example.

P1 Big Numbers (BN) - P12 [Score: 5]

Implement a calculator for large numbers based on P11 and any additional methods you define. The user first enters a number, then an operator, then the second number; print the result. The operator may be -, +, *, /, %, ^, or !. For !, a second number is not required.

P2 Polynomial Lab (PL) - P21 [Score: 5]

Implement PSum, PMinus, and PMultiple for two polynomials represented as strings.

P2 Polynomial Lab (PL) - P22 [Score: 5]

Based on P21 and any helper methods you add, implement PDiv, returning the quotient as a string (5 marks), and PMod, returning the remainder as a string (-5 marks).

P3 Math Lab (ML) - P31 [Score: 5]

ML31: receive as a string a mathematical expression containing integers and - and +, and return the result as a long. Examples: -438-2+23 or 329+9-9128.

P3 Math Lab (ML) - P32 [Score: 5]

ML32: write ML31 for the case where the numbers may also be decimal.

P3 Math Lab (ML) - P33 [Score: 5]

ML33: write ML32 for the case where parentheses are also added.

P3 Math Lab (ML) - P34 [Score: 10]

ML34: write ML33 for the case where *, /, and ^ are also added.

P3 Math Lab (ML) - P35 [Score: 10]

ML35: write ML34 for the case where functions such as Exp, Sqr, Ln, Log, Tan, Cos, and Sin may also appear. Missing each method carries a -3 mark penalty.

P4 ML2 - P41 [Score: 10]

Based on method P35, read a nonlinear equation as a string and print all of its roots.

P5 Files - P51 [Score: 30]

Restaurant information registration system. Required features: 1) Add, edit, delete, and perform simple and advanced searches on restaurant member data including membership number, first name, last name, national ID, phone number, and address; store all member data in a file sorted by membership number. (5 marks) 2) Add, edit, delete, and perform simple and advanced searches on types of foods and desserts, including food number, title, unit price, quantity available, and minimum stock threshold; store food data in a file sorted by food number. (5 marks) 3) Register sales invoices, search registered invoices, and deactivate invoices. (10 marks) 4) Display statistical data such as available foods, best-selling foods, members with the most purchases, today's cash-box balance, foods whose stock is below the threshold, etc. (10 marks)

P6 Quick Fact - P61 [Score: 5]

Calculate $100000!$, $200000!$, ..., $1000000!$ and store them in separate files.

P6 Quick Fact - P62 [Score: 5]

Read a number and, by reading the nearest stored value from a file, calculate its factorial and store it in a file named with the input number.

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First Semester 1393

P1 Big Numbers (BN) - P11 [Score: 5]

Implement BigSum, BigMinus (first number larger), BigMultiple, and BigCompare (returns 1, -1, or 0). Then implement BigDiv (5 marks) and BigMod (-5 marks), and show their use with an example.

P1 Big Numbers (BN) - P12 [Score: 5]

Implement a large-number calculator based on P11. The operator may be -, +, *, /, %, ^, or !; ! does not require a second number.

P2 Polynomial Lab (PL) - P21 [Score: 5]

Implement PSum, PMinus, and PMultiple for two polynomials represented as strings.

P2 Polynomial Lab (PL) - P22 [Score: 5]

Based on P21, implement PDiv (5 marks) and PMod (-5 marks).

P3 Math Lab (ML) - P31 [Score: 5]

ML31: receive a string expression containing integers and - and + and return a long. Examples: -438-2+23 or 329+9-9128.

P3 Math Lab (ML) - P32 [Score: 5]

ML32: extend ML31 so the numbers may be decimal.

P3 Math Lab (ML) - P33 [Score: 5]

ML33: extend ML32 by adding parentheses.

P3 Math Lab (ML) - P34 [Score: 10]

ML34: extend ML33 by adding *, /, and ^.

P3 Math Lab (ML) - P35 [Score: 10]

ML35: extend ML34 so it may include Exp, Sqr, Ln, Log, Tan, Cos, and Sin. At each stage, use and extend the methods from the previous stage.

P4 Big Bases (BB) - P41 [Score: 5]

Read a large number as a string and print its equivalents in base 16 and base 32.

P4 Big Bases (BB) - P42 [Score: 5]

Using an appropriate coding table, extend P41 to base 100.

P4 Big Bases (BB) - P43 [Score: 5]

Extend P42 to all bases less than 1,000,000.

P4 Big Bases (BB) - P44 [Score: 15]

Implement addition and multiplication of two large numbers in large bases (up to 1,000,000).

P5 Files - P51 [Score: 30]

Bank information registration system. Required features: 1) Add, edit, delete, and search bank-customer data including account number, balance, first name, last name, national ID, phone number, address, and account-opening date; store customer data in a file sorted by account number. (5 marks) 2) Register deposit information including deposit-document number, account number, date, national ID, depositor first and last name, and deposited amount; store all deposits in a separate file. (5 marks) 3) Register withdrawal information including withdrawal-document number, account number, date, and withdrawn amount; store all withdrawals in a separate file. (5 marks) 4) Block and Unblock a specific account (preventing withdrawal), and support deletion of a withdrawal or deposit document (isDelete). (5 marks) 5) Display statistics of a specific account's banking transactions in a specified date range, sorted by date. (5 marks) 6) Display statistics of the bank's total balance (sum of bank accounts) over a specified date range. (5 marks)

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First Semester 1394

P1 Big Numbers (BN) - P11 [Score: 10]

First implement BigSum, BigMinus (first number larger), BigMultiple, and BigCompare (returns 1, -1, or 0). Based on those methods, implement BigDiv (5 marks) and BigMod (-5 marks), and write a simple test program for each. Then, using the written methods, obtain the integer result of: $1^{1/1!} - 2^{2/2!} + 3^{3/3!} - \dots + 999999^{999999/999!}$ (5 marks).

P1 Big Numbers (BN) - P12 [Score: 5]

Write a method that receives a positive large number as a string and returns its base-16 equivalent as a string. (5 marks) Then write a program that prints the series $1^1 + 2^2 + 3^3 + \dots + 100^{100}$ in base 16. (-5 marks)

P1 Big Numbers (BN) - P13 [Score: 10]

Calculate $(2^{64})!$ and store the result in a text file.

P1 Big Numbers (BN) - P14 [Score: 10]

Calculate the first prime number with one million digits and store the result in a text file.

P2 Polynomial Lab (PL) - P21 [Score: 5]

Implement PSum, PMinus, and PMultiple for two polynomials represented as strings.

P2 Polynomial Lab (PL) - P22 [Score: 5]

Based on P21, implement PDiv (5 marks) and PMod (-5 marks).

P3 Math Lab (ML) - P31 [Score: 5]

Write a method that receives as a string a mathematical expression containing ordinary decimal numbers and - and +, and returns the result as a decimal number. Example: $-438.001078 - 0.2021 + 2.3101003$.

P3 Math Lab (ML) - P32 [Score: 5]

Using ML31, add *, /, and ^ to the mathematical expression, respecting operator precedence.

P3 Math Lab (ML) - P33 [Score: 5]

Using ML32 and ML31, add parentheses to the mathematical expression, respecting operator precedence.

P3 Math Lab (ML) - P34 [Score: 10]

Using the previous methods, write a method named ML that receives a complete mathematical expression containing all previous operators and functions such as Exp, Sqr, Ln, Log, Tan, Cos, and Sin, and returns the result as a decimal number.

P4 Numerical Analysis Project (NAP) - P41 [Score: 5]

Read an integer n from the first line of a text file named m.txt. Then read an n x n matrix of decimal numbers from the same file. Calculate the inverse and store it in m-Inserve.txt (filename preserved as written in the source).

P4 Numerical Analysis Project (NAP) - P42 [Score: 5]

Read an integer n from the first line of a text file named m1.txt. Then read an n x (n+1) matrix of decimal numbers from the same file as the coefficients of a linear system. Solve the system using Gauss or Gauss-Jordan and store the answers in m1-Gauss.txt and m1-Jordan.txt.

P5 Files - P51 [Score: 20]

Room accounting information registration system. Required features: 1) Member registration: add, edit, delete, and search member data including a unique member number (or student number), first name, last name, phone number, and registration date; store all member data in one file sorted by member number. (5 marks) 2) Financial expense-document registration: record a unique document number, full description of items and reasons for the expense, paying member number, payment date, total paid amount, and the members covered by the expense; store all expense data in a separate file. (10 marks) 3) Display the sum of all expenses in sorted tabular form. (5 marks)

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Section 7 - Sample Final-Exam Questions

First Semester 1391

Question 1 [Score: 10]

Write a program that reads 100 numbers and prints the count of numbers that both occur in the Fibonacci sequence and are greater than the average of the 100 input numbers. Explain the different parts of the written program in Persian. Failure to do so: -5 marks.

Question 2 [Score: 10]

Write a program that calculates and prints the series $1 - 2^2/3! + 3^3/5! - 4^4/7! + 5^5/9! - \dots - 100^{100}/199!$ with accuracy 0.0000001. The program must not produce an Error. Explain the different parts of the written program in Persian. Failure to do so: -5 marks.

Question 3 [Score: 10]

Write a method that receives strings S1 and S2, finds and removes every S2 inside S1, and returns the result. Explain the different parts of the written program in Persian. Failure to do so: -5 marks.

Question 4 [Score: 10]

Write a program that reads a 10 x 10 matrix from a file named A.txt, sorts each column of the matrix, and stores the resulting matrix in B.txt. Explain the different parts of the written program in Persian. Failure to do so: -5 marks.

Question 5 [Score: 10]

Write a program that reads 100 sorted numbers into an array, then reads a number M and searches for it. The search must be like binary search except that at each stage the array is divided into 3 parts; after comparison, 2 of the 3 parts are discarded and the process continues on the remaining part (ternary search). Explain the different parts of the program and the variables used in Persian. Failure to do so: -5 marks.

Question 6 [Score: 25]

Write a program that reads as a string a mathematical expression containing +, multiplication, parentheses, and positive numbers, then prints the result. You may define and use different methods for different tasks (+10 marks). Explain the different parts of the program and variables used in Persian. Failure to do so: -10 marks.

Question 7 [Score: 25]

Assume methods BigDiv, BigMod, BigSum, BigMinus, BigMultiple, and BigCompare exist, with BigCompare returning 1, -1, or 0. Write a program that reads one large number and factorizes it into prime factors. You may define additional methods. Explain the different parts of the program and variables used in Persian. Failure to do so: -10 marks. Important: the numbers and exponents obtained from the prime factorization may themselves be large numbers.

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Second Semester 1391

Question 1 [Score: 10]

Write a program that reads a number and prints the count of its divisors that are greater than the average of all divisors of that number.

Question 2 [Score: 10]

Write a program that calculates and prints the series $1!/1^1 - 3!/3^3 + 5!/5^5 - 7!/7^7 + \dots$ with accuracy 0.0000001. The program must not produce an Error.

Question 3 [Score: 10]

Write a program that calculates and prints all roots of a specific nonlinear equation such as $x^3 - x^2 - 6x + 3$.

Question 4 [Score: 10]

4-1) Write a method that receives a string, capitalizes the first letter of every word, and returns the result. (5 marks) 4-2) Write a method that receives a string, separates its words, and returns them as an array of strings. (5 marks)

Question 5 [Score: 10]

Write a program that reads a number in base 10, calculates its equivalent in base 32, and prints it.

Question 6 [Score: 10]

Write a program that reads 1000 numbers from a file named A.txt, sorts them, and stores the result in B.txt.

Question 7 [Score: 40]

Assume methods BigDiv, BigMod, BigSum, BigMinus, BigMultiple, and BigCompare exist. 7-1) Print the result of $1^1 + 2^2 + 3^3 + \dots + 100^{100}$. (10 marks) 7-2) Calculate and print the 100000th Fibonacci number. (10 marks) 7-3) Calculate and print the 100000th prime number. (20 marks)

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First Semester 1390**Question 1 [Score: 10]**

Write a program that reads 50 numbers and prints the average of the values that are both prime and present in the Fibonacci sequence. Explain the different parts of the written program in Persian. Failure to do so: -5 marks.

Question 2 [Score: 15]

Write a program that calculates and prints the series $1 + 2!/2^2 + 3!/3^3 + 4!/4^4 + \dots$ with accuracy 0.0000001. The program must not produce an Error. Explain the different parts of the written program in Persian. Failure to do so: -5 marks.

Question 3 [Score: 10]

Write a method that receives a string, converts its lowercase letters to uppercase and its uppercase letters to lowercase, and returns the result. Explain the different parts of the written program in Persian. Failure to do so: -5 marks.

Question 4 [Score: 15]

Write a program that reads a 10 x 10 matrix from A.txt, sorts each row of the matrix, and stores the resulting matrix in B.txt. Explain the different parts of the written program in Persian. Failure to do so: -5 marks.

Question 5 [Score: 20]

Write a program that reads 100 numbers into an array, finds the numbers with the greatest frequency, and prints them together with that frequency. Several numbers may tie for the greatest frequency; the program must print all such values and their frequency. Explain the different parts of the program and variables used in Persian. Failure to do so: -5 marks.

Question 6 [Score: 30]

Write a program that reads as a string a mathematical expression containing +, -, *, /, parentheses, and numbers that may have multiple digits, then calculates and prints the result. Explain the different parts of the program and variables used in Persian. Failure to do so: -10 marks.

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First Semester 1392**Question 1 [Score: 10]**

Write a program that reads a number, calculates and prints the average of its prime divisors. (5 marks) Fully test the program for the number 10. (5 marks)

Question 2 [Score: 10]

Write a program that reads 10 numbers into an array, then reads n and circularly shifts the array n positions forward; print the resulting array. (5 marks) Fully test the program with 3, 4, 65, 33, 12, 7, 44, -5, 4, 8 and n=3. (5 marks)

Question 3 [Score: 10]

Write a method that receives a string, lowercases all words, capitalizes only the first letter of each word, and returns the result. (5 marks) Fully test the method for "aLi iS a tEAchEr". (5 marks)

Question 4 [Score: 10]

Write a program that reads a base-10 number, calculates and prints its equivalent in base 32. (5 marks) Fully test the program for 8345. (5 marks)

Question 5 [Score: 10]

Write a program that reads a 10 x 10 matrix from the text file D:\A.txt, sorts its columns, and stores the result in D:\B.txt.

Question 6 [Score: 10]

Write a method that receives a string containing an ordinary decimal number such as "-328.098" or "101.009", converts the decimal value in the string to a decimal number, and returns it. (5 marks) Fully test the method for "-430.008". (5 marks)

Question 7 [Score: 10]

Answer only one of these: 7-1) Read a string of at most 10 characters and print all its subsets. 7-2) Read a string of at most 10 characters and print all its permutations.

Question 8 [Score: 30]

Assume methods BigDiv, BigMod, BigSum, BigMinus, BigMultiple, and BigCompare exist. 8-1) Print the result of $1^1 + 2^2 + 3^3 + \dots + 100^{100}$. (10 marks) 8-2) Calculate and print the (2^{100}) th prime number. (20 marks) Note: 2^{100} does not fit in a long or double.

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First Semester 1393**Question 1 [Score: 10]**

Write a program that reads a number and prints the digits of that number that are smaller than the average of the digits of the same number. (5 marks) Fully test the program for 43113271. (5 marks)

Question 2 [Score: 10]

Write a program that reads 10 numbers into an array, then reads n and removes from the array every value whose sum of divisors equals n. Removal must be performed by shifting values and reducing the array length. Print the remaining values. (5 marks) Fully and step-by-step test the program with 9, 3, 7, 5, 6, 3, 8 and n=6. (5 marks)

Question 3 [Score: 10]

Write a method that receives a string, removes its repeated characters, and returns the result. (5 marks) Fully test the method for "aLi iS a tEAchEr". (5 marks)

Question 4 [Score: 10]

First write a method named FullTrim that receives a string, removes all extra spaces, and returns the result. (5 marks) Then write a program that reads a multi-line collection of Strings from D:\A.txt, uses the previous method to remove all extra spaces, and stores the result in D:\B.txt. (5 marks)

Question 5 [Score: 10]

Write a program that reads n, then reads n numbers into an array. The entered numbers have a special order: from position i onward, all values in the array are sorted in descending order. After reading and storing the n numbers, find and print i.

Question 6 [Score: 15]

Write a program that reads strings S1 and S2 and prints the words common to both strings. Before writing the program, identify all required methods, name them, explain them one by one, and implement them. Finally write the requested program using the methods you created.

Question 7 [Score: 15]

Write a program that reads a sentence and removes repeated words. The instructions of Question 6 also apply to this question.

Question 8 [Score: 20]

Assume methods BigDiv, BigMod, BigSum, BigMinus, BigMultiple, and BigCompare exist. Write a program that calculates and prints the (2^{100}) th prime number, explaining every step of the program. Note: 2^{100} does not fit in a long or double.

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First Semester 1394**Question 1 [Score: 10]**

Write a program that reads a number and prints the divisors of that number that occur in the Fibonacci sequence. (5 marks) Fully test the program for 16. (5 marks)

Question 2 [Score: 10]

Write a program that reads a polynomial as a string, finds its greatest exponent, and prints it.

Question 3 [Score: 10]

Write a program that reads as a string a mathematical expression containing ordinary decimal numbers and the - and + operators, and prints the result as a decimal number. Example: $-438.001078 - 0.2021 + 2.3101003$. You may use parseDouble.

Question 4 [Score: 10]

First write a method named FullTrim that receives a string, removes all extra spaces, and returns the result. (5 marks) Then write a program that reads a multi-line collection of Strings from D:\A.txt, uses the previous method to remove all extra spaces, and stores the result in D:\B.txt. (5 marks)

Question 5 [Score: 10]

Write a method that receives a string, removes consecutive repeated words, and returns the result. A word is distinguished from the next word by the occurrence of the space character.

Question 6 [Score: 15]

Write a program that reads a number and prints its equivalent in base 32. Base-32 digits are:

0,1,2,3,4,5,6,7,8,9,A,B,C,...,V,10,... . Use a more efficient method; using many if statements will receive no credit.

Question 7 [Score: 15]

First write these methods: setRand receives a 20-element array and places random numbers from 1 to 100 in that same input array. (5 marks) isSort receives a 20-element array, checks whether it is sorted, and returns 0 or 1. (5 marks) Then use these methods to repeatedly generate random numbers from 1 to 100 in the 20-element array and check whether the array is sorted; repeat until the generated random values are sorted. (5 marks)

Question 8 [Score: 20]

Assume BigSum, BigMinus, BigMultiple, BigCompare, BigDiv, and BigMod have all been written. Write a program that calculates and prints the first 1000-digit prime number.

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